

LAST NAME: _____

Practice Quiz 08B — MATH 2055 — Fall 2026

Practice for Quiz 08

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Auto-generated by the Codex harness using GPT-5.

Recall that $u \equiv v \pmod{4}$ means $4 \mid (u - v)$.

1. Define a relation on $\mathbf{Z}$ by $x \sim y$ if and only if $x \equiv y \pmod{4}$.
 - (a) Prove directly from the definition that $\sim$ is transitive.
 - (b) Assume that $\sim$ is also reflexive and symmetric. How many equivalence classes are there? List them.